Name: S.HEMAA

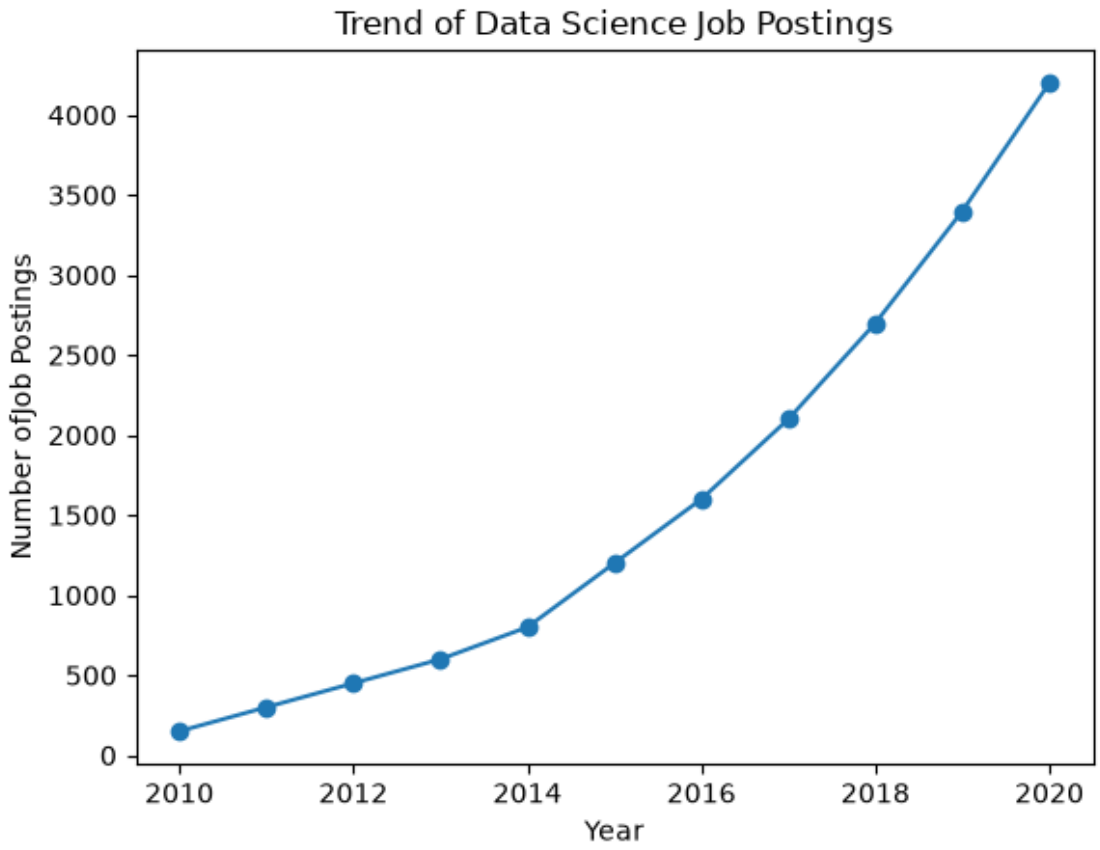
Register No: 250701238

Experiment name: Data Science Job Analysis, Data Classification and Data Security

1.a)

```
import pandas as pd
import matplotlib.pyplot as plt
data = {'Year': list(range(2010, 2021)),
'Job Postings': [150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400,  4200]}

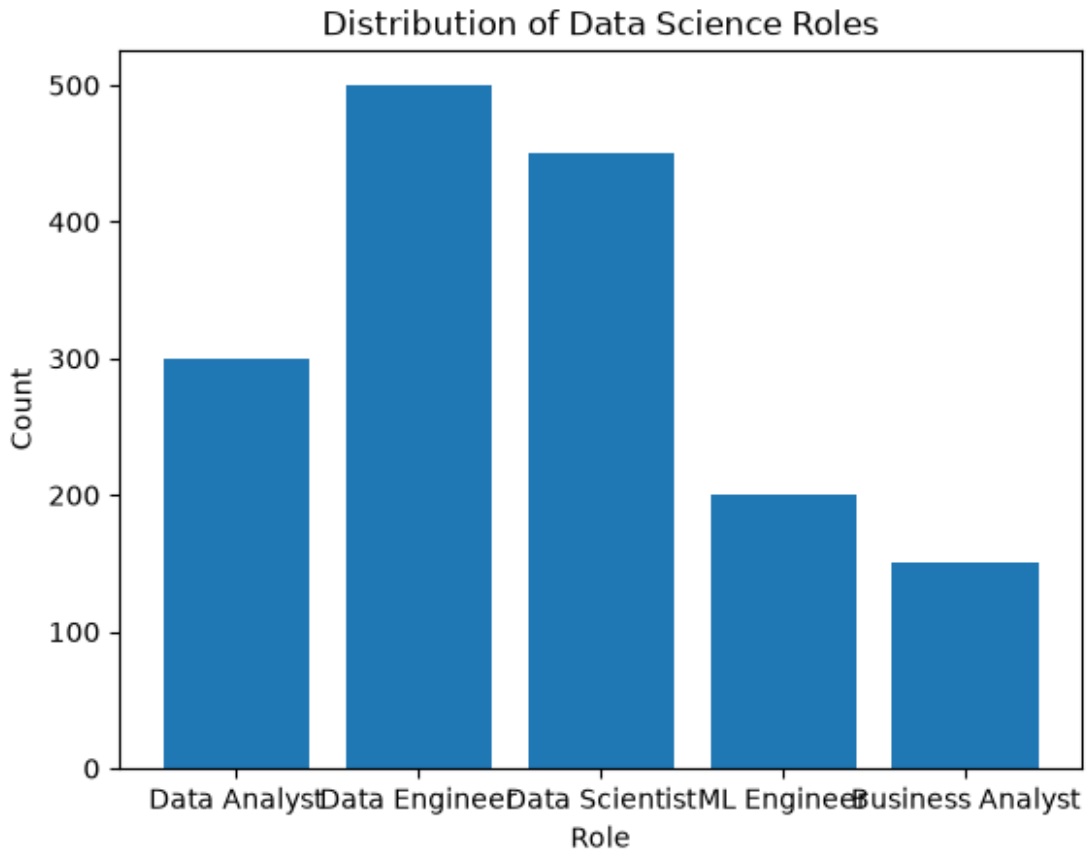
df = pd.DataFrame(data)
plt.plot(df['Year'], df['Job Postings'], marker='o')
plt.title('Trend of Data Science Job Postings')
plt.xlabel('Year')
plt.ylabel('Number ofJob Postings')
plt.show()
```



1.b)

```
roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer',
'Business Analyst']
```

```
counts = [300, 500, 450, 200, 150]
plt.bar(roles, counts)
plt.title('Distribution of Data Science Roles')
plt.xlabel('Role')
plt.ylabel('Count')
plt.show()
```



3.c)

```
import pandas as pd
structured_data=pd.DataFrame({"Name":["Rahul","Priya","Arun"],"Age":[21,22,20
],"Department":["CSE","IT","ECE"]})
print("STRUCTURED DATA")
print(structured_data)
print("\nSEMI-STRUCTURED DATA")
semi_structured={"Student":{"Name":"Rahul","Age":21,"Skills":["Python","SQL",
"Power BI"]}}
print(semi_structured)
print("\nUNSTRUCTURED DATA\n")
print("Data Science is one of the fastest growing fields.")
print("It is used in healthcare,banking,education and many other
industries.")
```

```
STRUCTURED DATA
    Name  Age Department
```

```
0  Rahul   21        CSE
1  Priya   22         IT
2   Arun   20        ECE

SEMI-STRUCTURED DATA
{'Student': {'Name': 'Rahul', 'Age': 21, 'Skills': ['Python', 'SQL', 'Power
BI']}}

UNSTRUCTURED DATA

Data Science is one of the fastest growing fields.
It is used in healthcare,banking,education and many other industries.
```

1.d)

```
# Generate key and encrypt data
from cryptography.fernet import Fernet
key = Fernet.generate_key()
f = Fernet(key)
token = f.encrypt(b"Rajalakshmi Engineering College")
token
b'...'
f.decrypt(token)
b'Rajalakshmi Engineering College'
key = Fernet.generate_key()
cipher_suite = Fernet(key)
plain_text = b"Rajalakshmi Engineering College."
cipher_text = cipher_suite.encrypt(plain_text)
# Decrypt data
decrypted_text = cipher_suite.decrypt(cipher_text)
print("Original Data:", plain_text)
print("Encrypted Data:", cipher_text)
print("Decrypted Data:", decrypted_text)
```

```
Original Data: b'Rajalakshmi Engineering College.'
Encrypted Data: b'gAAAAABqfe1TSrIzTUG4AK2BcvUm7gzuZOqLz-
6YNydBWuxzW6yQHUJUsuKQRyrE8yYkfCotqHXE9kQ9N_2Jomob5FeSRBkPpxPDljhOkTj-qw-
5TlMs_xgStjqF-5k73F5qUA3Ohy4R'
Decrypted Data: b'Rajalakshmi Engineering College.'
```